

Charotar University of Science and Technology

M.Sc. (Nano Science and Technology) Semester II Examination April 2010

NT 708 Concepts of Chemical and physical Sciences II

Date: 28th April 2010

Time: 10.00 to 10.30 am

Max. Marks 20

Part-A

Do as directed (Show your calculation within the space provided)

1. In a certain process, 600 J of heat is absorbed by a system, while 290 J of work is done on the system. What is the change in internal energy for the process? (2)

(a) 890 J (b) 310 J (c) 290 J (d) 600 J
2. For an ideal gas $C_p - C_v$ is (2)

(a) R (b) $\frac{1}{2}R$ (c) -R (d) $\frac{3}{2}R$
3. Identify the reaction order from the following value of rate constant (2)
 $k = 5.6 \times 10^{-4} \text{ mol.dm}^{-3} \text{ s}^{-1}$

(a) second (b) zero (c) first (d) third
4. Time required for the half completion of a reaction is found to be independent of the initial concentration of reactants. The order of the reaction is: (2)

(a) zero (b) second (c) first (d) third
5. Hybridization of carbon in benzene is: (1)

(a) sp (b) sp^3 (c) sp^2 (d) none of these
6. In radio carbon dating, the isotope of carbon used is: (1)

(a) C^{12} (b) C^{13} (c) C^{14} (d) none of these

7. The CH bond length in methane is: (2)

- (a) less than that of CH in ethylene
- (b) greater than that of CH in ethylene
- (c) equal to CH bond in ethylene

Justify your answer.

8. In sp^3 hybridized carbon the $\angle HCH$ is (2)

- (a) 120°
- (b) 109.5°
- (c) 180°
- (d) none of these

9. The non colligative properties are (1)

- (a) Elevation of boiling point
- (b) Osmotic pressure
- (c) Depression of freezing point
- (d) Ionization of a strong base

10. Given $R = 8.206 \times 10^{-2} \text{ dm}^3 \text{ atm. K}^{-1} \text{ mol}^{-1}$ and $R = 8.1344 \text{ J K}^{-1} \text{ mol}^{-1}$ Complete the following (show the calculation) (2)

$\text{atm dm}^3 = \text{-----}$

11. One mole of an ideal monoatomic gas is taken from the state ($v = 22.4 \text{ lit}$, $p = 1 \text{ atm}$, $\text{temp} = 273^\circ \text{ K}$) to the state ($p = 2 \text{ atm}$, $\text{temp} = 303^\circ \text{ K}$). The ΔU is : (show the calculation and tick the correct answer) (3)

- (a) 90 cal
- (b) 150 cal
- (c) 100 cal
- (d) none of these

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Max. Marks 50

Part-B

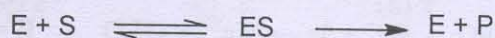
(Write each section on separate answer book)

SECTION-I

- Que.1 (a) A first order reaction is 25 % complete at the end of 20 min. How long will it take to complete 75 % of the reaction? (3)
- (b) The following consecutive reaction $A \rightarrow B \rightarrow C$ involves all first order reaction. Derive a relation for $T_{\max}$, when the concentration of B is maximum; show that the slowest step determines the rate. (7)

OR

- Que.1 (a) For a reaction $A \rightarrow B + C$, it is found that the rate increases by a factor of 2.25, when concentration is increased by a factor of 1.5 at the same temp. What is the order of the reaction? (3)
- (b). It is assumed that the enzyme catalysis occurs via rapid and reversible formation of complex between enzyme(E) and substrate(S). The complex (ES) breaks down to give product and regenerate enzyme. The reactions are:



Derive Michaelis –Menten equation. Explain the significance of Michaelis constant (K_M). Discuss in brief how K_M may be obtained experimentally. (7)

- Que.2 (a) State Raoult's law. (2)
- (b) 0.5 g of a nonvolatile solute of molar mass 60 g mol^{-1} is dissolved in 100 g of ethyl acetate at 20°C . What will be the vapor pressure of the solution at 20°C ? The vapor pressure of ethyl acetate at 20°C is 72.8 torr. (3)
- (c) Explain "Reverse Osmosis" in short. (2)
- (d) A solution containing 1.6 g per dm^3 of a nonvolatile solute has an osmotic pressure of 8.3 torr at 25°C . Calculate the molar mass of the solute. $R=0.0821 \text{ dm}^3\text{atm K}^{-1}\text{mol}^{-1}$. (3)

SECTION-II

- Que.3 (a) Calculate the change in entropy, when one mole of He gas, assumed to be ideal is heated from a temperature of 298°K to a temperature of 400°K at constant pressure. Given: $R=8.314 \text{ JK}^{-1}\text{mol}^{-1}$. (3)
- (b). Calculate the change in entropy, when 5 g tin are heated from 20°C to 300°C . Given; per gram of tin the latent heat of fusion is 14 cal, sp. heats of solid and liquid tin are 0.55 and 0.064 respectively. Melting point of tin is 232°C . (7)

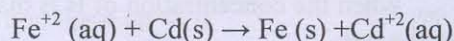
OR

- Que. 3 (a) 5 moles of an ideal gas was expanded isothermally and reversibly from a pressure of 10 atm to 2 atm at 300 °C. What is the largest mass that can be lifted through a height of 1 meter in this expansion? $g = 9.81 \text{ ms}^{-2}$. (4)
- (b) 100 g of ice at 0 °C is dropped into an insulated beaker containing 150 g of water at 100 °C. Calculate the change in entropy for the process that then occurs. Heat capacity of ice = 0.5 cal/ deg-g. Latent heat of fusion of ice = 80 cal/ g. (6)

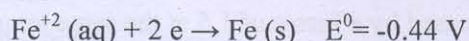
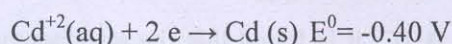
- Que. 4 (a). A Zn rod is placed in a 0.1 M solution of ZnSO_4 at 25 °C. Assuming that the salt is dissociated to the extent of 95 % at this dilution, calculate the electrode potential at these temperature. Given $E^0 = -0.76 \text{ V}$ for the reaction: (3)



- (b). Set up the electrochemical cell that involves the reaction: (4)



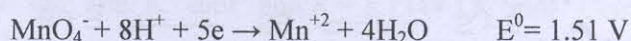
and calculate e.m.f of the cell. The half cell reactions are:



- (c). Obtain the mass of copper produced by the passage of 1.60 amperes of current through a solution of copper sulfate for 1.00 hr. Given: 1 Faraday = 96487 coulombs. (3)

OR

- (c). The following half reactions are given: (3)



With proper explanation, indicate which one of the following statements is correct.

- Chloride ion is oxidized by permanganate.
- Permanganate is oxidized by chloride
- Cl^- and permanganate will not undergo any reaction.

- Que. 5 (a). Discuss the relative stabilities of tertiary, secondary and primary carbonations. (5)
- (b). Explain with suitable examples, how does intra -molecular hydrogen bonding influence strength of organic acids. (5)